

SPACE INVADERS ARDUINO CONTROLLER

WHAT IS THE PROJECT CONCEPT?

A Space Invaders arcade game controlled by physical Arduino hardware instead of a keyboard. Players use two potentiometers and two buttons connected to an Arduino to control a spaceship. The project demonstrates real-time serial communication between Arduino sensors and a p5.js web game, combining classic arcade gameplay with tangible physical computing.

HOW DO YOU USE IT?

Controls: - Potentiometer 1: Move ship up/down - Potentiometer 2: Aim shooting angle (-60° to +60°) - Left Button: Move ship left - Right Button: Move ship right - Auto-fire continuously based on aim

Setup: 1. Wire Arduino with 2 potentiometers (analog pins A12, A10) and 2 buttons (digital pins 13, 14) 2. Upload Arduino code that sends comma-separated sensor values via serial 3. Run p5.serialcontrol app 4. Open p5.js sketch in browser 5. Destroy all enemies before they reach the bottom to win

HOW IS IT WORKING TECHNICALLY?

System Architecture: Arduino reads analog potentiometers (0-255) and digital buttons (0/1) every 2ms, then sends data via UART serial at 9600 baud to p5.serialcontrol. The p5.js sketch receives this data through the p5.serialport library and updates game state in real-time.

Communication: Arduino sends: "potPos,potAim,btnLeft,btnRight" (example: "127,200,0,1") p5.js sends back: score value for optional LED display

Technical Implementation: The Arduino uses ADC (Analog-to-Digital Conversion) to read potentiometer voltages and convert them to 8-bit values (0-255). Digital pins read button states. Data is transmitted via UART serial protocol.

The p5.js game runs at 60fps with object-oriented classes for Player, Bullet, Enemy, and Particle objects. It uses p5.Vector for physics calculations including position, velocity, and acceleration. The serialEvent() callback parses incoming Arduino data and maps sensor values to game coordinates using the map() function.

Input mapping converts potentiometer values to screen positions and angles. Collision detection uses Euclidean distance calculation. Enemy movement follows classic Space Invaders AI (horizontal sweep, reverse at edges, descend). Particle explosions simulate physics with gravity and random velocity vectors.

Data flows from Arduino sensors through serial to p5.serialcontrol to the browser, where the game updates player position/aim, fires bullets at cooldown intervals, detects collisions, updates particle effects, and sends score feedback back to Arduino.

Technologies: Arduino (C/C++), p5.js (JavaScript), p5.serialport library, UART serial communication, vector mathematics, event-driven programming